

Not Just an Evaluation: A Case Study of Discourse Particle Sensitivity in Language Models

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Abstract

Discourse particles subtly shape the flow of discourse by reflecting conversational context and interlocutor attitudes, often leaving the truth-conditional meaning untouched. They are also often polyfunctional, with distinct senses which partially overlap with other particles; for example, the exclusive use of English *just* and *only* (John *only*/just drinks water). Despite the frequency of these words in natural language, they have gone largely unstudied due to the subtle nature of their effects. We investigate LLM sensitivity to *just* and *only* in natural conversations using two behavioral experiments and a localization probe. First, we find that LLMs and humans similarly prefer the presence of these particles in context, and both are more sensitive to *just* compared to *only*. Second, when sampling from LLMs with and without these particles, we find LLMs consistently rate generations as more likely with their source context, demonstrating particle sensitivity in LLMs with their own distributions. In both experiments, instruction tuning increases sensitivity. Finally, we localize particle-sensitive units within LLMs and find the overlap between *just* and *only* is asymmetric: *only* units transfer to selectively to exclusive *just*, while all *just* units transfer to *only* equally well regardless of sense, suggesting LLM lexical-level particle representations that are partially organized by sense.

1 Introduction

While waiting for Bob to arrive at the theater, you and Claire discuss his potential tardiness:

You: Are you mad at him?

Claire: He’s (just) late all the time!

Whether or not Claire uses *just* in her response does not change Bob’s habitual tardiness; rather, it subtly shapes the future discourse: without *just*, you can easily question further (e.g. “Oh does this

happen often?”), while with *just*, further inquiry seems to push back on Claire’s claim that Bob is “just late”. In this dialog, *just* is an example of a **discourse particle**: a word that subtly shapes the flow of discourse and reflects conversational context and interlocutor attitudes while leaving the truth-conditional meaning largely untouched (Grosz, 2020). Hence, in evaluating sensitivity of these particles, one must consider their effects from prior context and on future context.

Another noteworthy aspect of discourse particles is that many are polyfunctional, having multiple distinct yet interconnected meanings which often partially overlap with other particles forming constellations of meaning (Aijmer, 2002). One such “constellation” is the group of *exclusive* particles (e.g. *only*, *just*, *solely*, *simply*): for example, Claire saying “I *only/just*¹ like popcorn” excludes other salient movie snacks from her liking (Rooth, 1992). Note that, while these particles share some uses, they do not completely overlap: Claire responding “He’s *only* late all the time.” in the dialog above is difficult to parse in some dialects and does not replicate *just*. Exclusive particles have been studied extensively in Semantics (Horn, 1969; Beaver and Clark, 2008); Coppock and Beaver (2014) propose a unified account for their meaning, a common core denotation for all exclusive particles.

Given the rise of Large Language Models (LLMs) and their acquisition of a variety of linguistic phenomena as tested in minimal-pair settings (Linzen et al., 2016; Marvin and Linzen, 2018; Warstadt et al., 2020, *i.a.*), it is natural to ask if they have also acquired sensitivity to these particles. This is further underscored by the frequency of discourse particles in conversation, a domain of ever-increasing importance for LLMs (especially as conversational agents). Finally, LMs are perhaps our strongest model of language *use*, so testing them on

¹The exclusive reading is one of several available for “just”

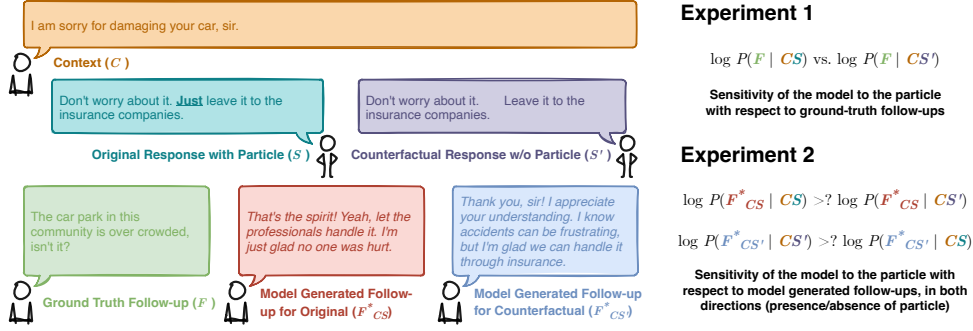


Figure 1: An example stimulus from our data, involving the context C , target sentence with particle S and counterfactual sentence without particle S' , ground-truth follow-up F , and LLM follow-ups generated with the particle F_{CS}^* and without $F_{CS'}^*$.

polyfunctional discourse particles in terms of how they affect upcoming utterances can give us empirical evidence about their theoretically-discussed effects, bridging gap between LLMs and Linguistic theory (Warstadt and Bowman, 2024; Futrell and Mahowald, 2026; Misra and Kim, 2026).

To this end, we investigate sensitivity of LMs to selected English discourse particles *just* and *only* using two behavioral experiments, including a comparison to human preferences. We then probe models for evidence of a unified representation for *exclusives*. We extract occurrences of our target words and their surrounding context from the natural conversations of DailyDialog (Li et al., 2017)—see Figure 1 for an example. We observe that 1) LLMs, especially instruct-tuned ones, and humans similarly prefer the ground-truth conversations with the particle over counterfactual versions with the particle removed (Section 4); 2) sampled follow-ups are consistently more probable under their source context (with/without the target words), showing particle sensitivity within models’ distributions (Section 5); 3) LLMs localize these particles lexically, and that some of this representation is shared between *only* and *exclusive just* in models’ residual layer, though its effects are largely asymmetric between the two.

This work builds ground for future explorations into pragmatically-sensitive continuations, which, to our knowledge, has gone understudied at the level of *discourse particles*. Work exists for high-level pragmatic relations (e.g. irony, metaphor; Hu et al., 2023), implicature (Ruis et al., 2023), discourse connectives (Pandia et al., 2021; Brubaker et al., 2026), and particle sense identification (Sheffield et al., 2025) or particle prediction (Yusoff et al., 2026), but none of these touch on the effect of discourse particles on continuations.

2 Background

Discourse Particles Discourse particles have been extensively studied in Linguistics (Grosz, 2020). The two particles we investigate, *just* and *only*, each have substantial lines of work. For *just*, several works analyze some of its senses and their relations (Lee, 1987; Beltrama, 2022; Wiegand, 2018; Warstadt, 2020; Windhearn, 2021), while Coppock and Beaver (2014) address *exclusive just* along with other exclusive particles, and Deo and Thomas (2025) propose a unified theory for 12 distinct senses. *Only* was first analyzed alongside *even* by Horn (1969), and was further studied with other exclusive particles by Beaver and Clark (2008) and Coppock and Beaver (2014).

Most of these works either (1) unify multiple senses of a single particle into one core account or (2) unify a use shared by multiple particles into a common core; our paper draws especially from Warstadt (2020), who proposes the core effect of *just* is to shape future discourse moves by communicating what the speaker views as unanswerable, and Coppock and Beaver (2014), who propose a unified account for exclusive particles.

Language Models & Discourse The study of discourse particles using LLMs is quite new compared to the extensive work done with the related discourse connectives (e.g. *however*, *since*, etc.). Connectives differ from discourse particle by signaling relations between clauses rather than speaker attitudes (Fraser, 1999). Pandia et al. (2021) use cloze-style tests to investigate LM capabilities at predicting connectives, and Brubaker et al. (2026) study LM capability to draw inferences from different connectives. Outside of discourse connectives, Schuster and Linzen (2022) find LMs hallucinate new entities in the discourse by failing to resolve

indefinite NPs in contexts such as negation; Kim and Misra (2026) find models follow pragmatic structure by preferring continuations that are at-issue and are sensitive to cues that humans use to modulate at-issueness; Hu et al. (2023) study LM performance on identifying high-level pragmatic relations such as irony and metaphor, and what linguistic cues LMs use to do so; and Ruis et al. (2023) study LM reasoning on implicatures that answer binary questions. Prior work that addresses discourse particles includes Sheffield et al. (2025), who focus on LM capability to distinguish different senses of discourse particles in a metalinguistic setting, and Yusoff et al. (2026), who prompt LMs to predict discourse particles in Colloquial Malay.

We draw from the minimal pair paradigm of Linzen et al. (2016) and BLiMP (Warstadt et al., 2020) and extend beyond the sentence to include pre- and post-context, following the theory of Warstadt (2020), for our first experiment (Section 4). Further, our second experiment (Section 5) takes inspiration from the paradigm of Kim and Misra (2026) where we compare generated continuations based on minimally differing contexts and applies it to study particle sensitivity rather than at-issue sensitivity.

Function Localization For our third experiment (Section 6), we adapt methods from activation patching that localize model components that show the greatest sensitivity to these particles. Prior work has used variations of these techniques to: localize mechanisms for subject–verb agreement (Finlayson et al., 2021); measure recovery of the factual prediction (Meng et al., 2022); localize regions that identify grammatical and ungrammatical features (Kryvosheieva et al., 2026). Our work extends beyond grammatical features to the pragmatics of discourse particles, and differs in methodology from Kryvosheieva et al. (2026) by instead localizing whole attention, MLP, and residual-stream activations at particular layers and evaluating them with counterfactual activation replacement.

3 Experimental Setup

3.1 Dataset

For all our experiments, we use a subset of conversations from DailyDialog (Li et al., 2017)², a set of natural dialogues, that contain our target words.

²DailyDialog is released with a CC0 1.0 Universal License and is free to use with attribution to Li et al. (2017)

We extract samples consisting of conversation context C , original utterance containing a target word S , a minimally edited utterance S' with the target word removed, and the ground-truth follow-up F (Figure 1). To help ensure the target word is relevant to the follow up in the conversation, we retain only examples where the target word appears in the final sentence of S and the target utterance is not a question. We additionally restrict the dataset to conversations whose context C contains at most three preceding turns, as preliminary experiments with longer contexts yielded noisier estimates of the subtle effects of word removal.

We filter for conversations where the utterance S contains discourse particles *just* or *only*. Additionally, since discourse particle effects are subtle and do not always alter the truth-conditional semantics of the sentence, we use *not* as a control since it has a clear impact on the truth-conditions. After filtering our dataset contains 1,182 *just* examples, 360 *only* examples, and 1,307 *not* examples.

From these data, we manually select 60 for each target word to form a gold dataset which is particularly sensitive to the presence/absence of the word.³ This provides a dataset which is both more manageable for some experiments and is predicted to provide the best signal. The *just* data are split by sense into two equal sense subsets, *exclusive* (“My picky brother just/only eats french fries and pizza”) and *non-exclusive*, for use in Section 6.

3.2 Scoring Setup

For probability scoring in our experiments, raw `__eu__` markers in DailyDialog conversations are converted into alternating Speaker 1:, Speaker 2: turn labels. Base models are scored directly on this speaker-labeled text, while instruction-tuned models are scored through their native chat templates using the same dialogue content. Following Hu and Levy (2023), we measure probabilities directly rather than prompt for metalinguistic judgments, as prompting can under-represent model knowledge. To avoid spurious penalties introduced by sentence-initial lowercasing after deletion, we capitalize the first alphabetic character of both utterance variants before scoring. Appendix A gives an end-to-end example of the C , S , S' , and F construction, both prompt renderings, and sampled follow-ups. Scoring metrics are specific to each experiment and discussed in corresponding sections.

³As judged by one of the authors, a native English speaker and linguist with background in semantics and pragmatics.

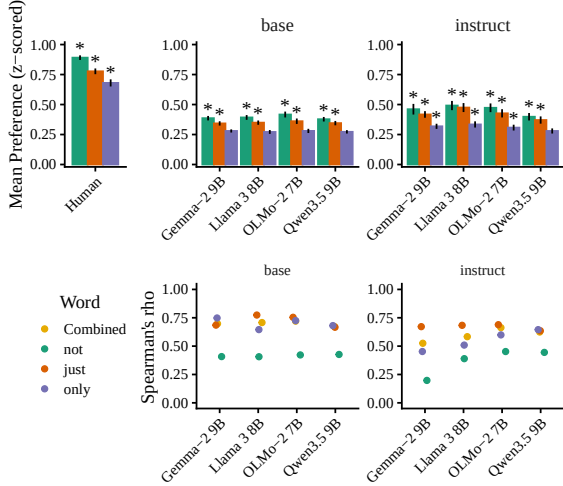


Figure 2: **Top:** Z-scored LM and Human preferences on the 180 gold data (distractors excluded). LM preference is defined in eq. (1), Human preference is rated on a 7-pt Likert scale, with 7 being strongly prefer the original passage with the word (CSF). Stars indicate significance.⁵ **Bottom:** Spearman’s ρ correlation between human and LM ratings, reported for both aggregate and each word.

3.3 Models

We study base and instruction-tuned checkpoints from four model families: Llama 3 8B (Grattafiori et al., 2024), OLMo-2 7B (OLMo et al., 2024), Qwen3.5 9B (Qwen Team, 2026), and Gemma-2 9B (Team et al., 2024). This model set keeps scale roughly comparable while spanning independently trained open-weight LLM families. Including both base and instruction-tuned checkpoints for each model family lets us compare discourse-particle sensitivity before and after post-training.

4 Exp 1: Particle Removal Effects for Humans and LMs

We begin investigating LLM sensitivity to discourse particles by comparing LM preferences to those of human participants. We measure LM and human preference between CSF and $CS'F$ on the gold dataset (Section 3.1), examining *just* and *only* with *not* as a meaning-altering control. Models that effectively encode these words’ conversational contribution should assign higher probability to the original dialogue than to the edited variant.

4.1 Scoring Metrics

LMs For models, our conversation-level score is the log-probability analogue of the comparison

introduced above:

$$\Delta_{\text{removal}} = \log P(CSF) - \log P(CS'F) \quad (1)$$

Positive values indicate that the model prefers the ground-truth conversation to the counterfactual one. Since S and S' have different token lengths, all reported deltas are normalized per token.

Humans In order to make the preference task for humans as similar to the task of the LMs, annotators are presented with CSF and $CS'F$ as continuous passages in two columns (randomized) and asked to rate their preference using an implicit 7-point Likert scale slider, which closely parallels the score computed for LMs. To establish baseline annotation validity, we included a set of distractor stimuli using randomly-sampled conversations from the dataset, not necessarily with one of our target words, where S' was formed by randomly reordering the words in S . The distractors were randomly mixed in with the actual data. In total, a set of 9 participants provided preferences for the gold data (240 total, 60 x 3 target words + 60 distractors), over two equal rounds of annotation. See Appendix B for the annotation interface and further details.

4.2 Results

All human participants consistently prefer the control distractor data (median rating 7), and high agreement across the dataset⁴. The top of Figure 2 shows LM and human preference ratings on the gold data. Humans show a preference for all words, and models follow this trend for *not* and *just*. The preferences for “only” are less strong for humans, and for all base models and one instruct model, the preference is not significantly above zero.⁵ We also observe generally strong positive correlations of models with humans across all words, and especially with *only* and *just*. The lower correlation with *not* appears to fall out of the concentrated distribution of human labels (they show strong, categorical-like preference) contrasted with the more graded LM ratings. Ultimately, we find humans and models qualitatively agree in their continuation preferences with respect to particles, with post-training increasing this sensitivity.

⁴Intraclass Correlation Coefficient with two-way random effects, absolute agreement, and averaged measures (Mcgraw and Wong, 1996) $ICC(A, k) = 0.854$, indicating high agreement (Koo and Li, 2016).

⁵greater-than-0 Wilcoxon signed-rank test with Bonferroni correction, 95%CI.

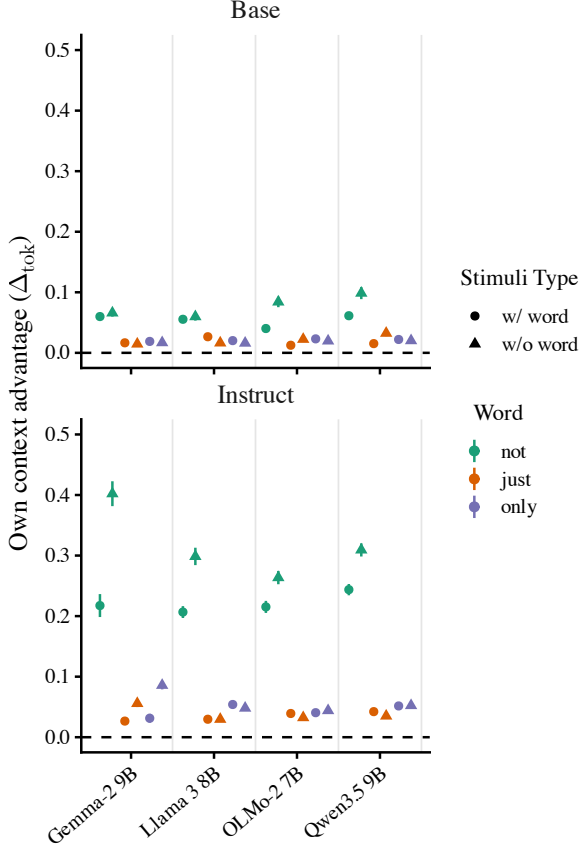


Figure 3: Own-context advantage by model in the generation experiment. We construct paired prompt contexts S and S' that differ only in the target word, sample next-turn responses from both prompts, and then rescore each sampled response under both contexts. Δ_{tok} is the mean per-token difference (Equation 2). Positive values indicate an own-context advantage.

5 Exp 2: Generation

Section 4 explored whether models penalize a minimally edited utterance when a target word is removed and compare to a ground-truth follow-up. This leaves two gaps: First, it paints an asymmetric picture of model sensitivity—we can only measure model preference for follow-ups that are appropriate only *with* the particle/word present. Second, we restrict the model to a single continuation, when models likely represent their own distribution of follow-up continuations (Kim and Misra, 2026). To address these, we turn to analyses that incorporate models own generated input, both in the presence and absence of a given particle/word. For each stimulus item, we compare paired prompt contexts CS and CS' that differ only in the presence of the target word, sample continuations from both versions, and then test whether each continuation is more probable under the context it came from than

under the alternate context. If a model represents the conversational contribution of a discourse particle, responses sampled from the particle-bearing prompt CS should show an own-context advantage relative to the particle-removed prompt CS' , and vice versa. We run this procedure for *just* and *only*, with *not* as a meaning-altering control.

5.1 Methodology

For each stimulus item (see Section 3.1), the sampling prompts are the observed conversation CS and its counterfactual CS' , where the target word is deleted from a naturally occurring utterance. For base models, prompts are rendered as speaker-labeled conversations ending at the next-speaker prefix. For instruction-tuned models, we use each tokenizer’s native chat template, placing the target utterance as the final user turn so that the model generates the next assistant response, without adding a task-specific system prompt (see Appendix A for a complete example).

From each of CS and CS' , we sample 50 unique next-turn responses using nucleus sampling (Holtzman et al., 2020) with $top_p = 0.9$ and temperature 1.0, truncating generations at the first speaker-turn boundary; prior work suggests that 50 samples offer a strong compute–performance tradeoff (Giulianelli et al., 2023). Each sampled response y is then rescored under both prompt variants. Let CS_{src} denote the prompt from which y was sampled and CS_{alt} the matched alternate prompt. We compute an own-context advantage:

$$\Delta_{\text{tok}}(y) = \bar{\ell}(y \mid CS_{\text{src}}) - \bar{\ell}(y \mid CS_{\text{alt}}), \quad (2)$$

where $\bar{\ell}(y \mid CS)$ is the sequence log-probability of y under prompt CS , normalized by the number of generated tokens. If y is sampled from CS , we compute $\bar{\ell}(y \mid CS) - \bar{\ell}(y \mid CS')$; if y is sampled from CS' , we reverse the subtraction so that positive Δ_{tok} values always favor the source context.

5.2 Results

Figure 3 shows that changing the target word changes the continuation space the model expects: in all model–word–source combinations, sampled responses are more probable under the prompt they were generated from than under the matched alternate prompt. As in Experiment 1, the largest effects come from *not*. Effects for discourse particles are roughly an order of magnitude smaller (Table 4) and, we see generally consistent preferences regardless of the edit. Overall, models do encode

discourse particles in next-turn expectations, but more subtly compared to a truth-conditional edit like removing *not*.

6 Exp 3: Function Localization Analysis

Experiment 2 showed that removing a discourse particle changes the continuations a model expects, but this behavioral effect does not reveal which internal computations support that sensitivity or whether models represent a shared structure across particles/words. Experiment 3 therefore asks whether the effect can be localized to particular layer–component sites and whether those sites are causally involved in preserving particle-sensitive response probabilities. We first localize sites for *not*, *just*, and *only* (Section 6.1), then test whether localization transfers specifically between the *exclusive sense of just* and the semantically similar particle *only* (Section 6.2).

6.1 Experiment 3a: Component-Based Localization of Word Sensitivity

Setup Our localization approach most directly adapts the *causal tracing* variant of Meng et al. (2022), using CS and CS' as minimal pairs. We apply counterfactual activation patching to the paired DailyDialog prefixes defined in Section 3.1. For each item, the clean condition is the observed prefix CS , comprising context C followed by the original utterance S containing the target word; the corrupted condition is the matched prefix CS' , in which S is replaced by its target-word-removed counterpart S' . In activation patching, an internal state produced by one input is substituted into a forward pass on another input, an intervention previously formulated as causal mediation and interchange intervention (Vig et al., 2020; Geiger et al., 2020; Heimersheim and Nanda, 2024). Since Llama 3 8B Instruct showed the highest average sensitivity in our generation experiments, we localize only on this model. At each of Llama 3’s 32 decoder layers, we treat each of the self-attention output, MLP output, and post-block residual stream as a separate candidate localization site.

Localization For each word, we begin with the responses generated in Section 5 from the CS prefixes of the 60 manually selected DailyDialog items described in Section 3.1; we retain 500 responses with the largest positive own-context advantage (Equation 2). These high-advantage responses form a pool for which the model’s sensitivity to

the target word is especially pronounced. We divide each pool into two folds by DailyDialog item, keeping all responses generated from the same CS prefix in the same fold. For each cross-validation split, localization is performed on one fold and causal evaluation on the other. To prevent items with many selected generations from dominating localization, we use only the highest-advantage response from each item when estimating site scores. This yields localization samples of 26 *not*, 39 *just*, and 32 *only* DailyDialog items, with one response per item.

For every response y in a localization fold and every candidate site s , we first run the model on CS and cache the full activation vector at site s for the final token of S , immediately before the chat template’s end-of-turn marker. We then run the matched CS' prefix and replace the activation at the same site for the final token of S' with the cached CS activation. Let $\bar{\ell}_{\text{patch},s}(y | CS')$ denote the per-token log-probability of y in this patched CS' run. We define the site’s restoration score as

$$R_s = \frac{\bar{\ell}_{\text{patch},s}(y | CS') - \bar{\ell}(y | CS')}{\max(\bar{\ell}(y | CS) - \bar{\ell}(y | CS'), 10^{-6})}.$$

Within each localization fold, we average R_s across DailyDialog items. We then rank the 32 layer-specific sites separately within each component type by mean restoration and retain the top $k \in \{1, 3, 5, 10\}$ sites.

Causal evaluation We evaluate the selected sites on the held-out fold using a counterfactual intervention to test for its necessity (Geiger et al., 2021; Heimersheim and Nanda, 2024). Starting from CS , we replace each selected site’s activation at the final token of S with the corresponding activation at the final token of S' from CS' and measure:

$$\Delta_{\text{patch}} = \bar{\ell}(y | CS; a_s \leftarrow a_s(CS')) \forall s \in \mathcal{S}_k - \bar{\ell}(y | CS).$$

A negative Δ_{patch} value indicates that replacing the representation of the observed utterance S with that of its target-word-removed counterpart S' lowers the response probability, providing causal evidence that the selected sites are necessary for the model’s word sensitivity. For each component and k , we compare the localized sites with four equally sized sets of randomly selected layers drawn from the remaining sites. Although localization uses only the highest-advantage response from each training-fold DailyDialog item, causal evaluation uses all retained responses from items in

the held-out fold. We therefore first average Δ_{patch} across responses from the same DailyDialog item and then across items within each fold. Table 1 reports the mean of the two fold-level effects.

Results As shown in Table 1, patching the localized sites has a significantly larger effect than patching random sites in 35 of the 36 word–component– k comparisons. The effect increases as more sites are patched, indicating that word sensitivity is both concentrated at highly informative sites and distributed across multiple layers. Residual-stream and attention sites produce the largest effects, while MLP effects are smaller. Across component types, the effects are strongest for *not*, followed by *only* and then *just*, matching the own-context ordering for Llama 3 8B Instruct in Experiment 2. These held-out interventions therefore provide causal evidence that the localized sites support the model’s word-sensitive response probabilities.

6.2 Experiment 3b: Sense-Specific Transfer Between *just* and *only*

The *exclusive* sense of *just* is semantically close to *only*. For example, in “She ate *just* rice” and “She ate *only* rice,” both particles exclude salient alternatives to rice. This shared alternative-excluding contribution has motivated unified semantic accounts of *exclusive* particles (Coppock and Beaver, 2014). *Non-exclusive* uses of *just* need not contribute the same meaning. This contrast lets us ask whether the component–layer sites identified in Experiment 3a encode the lexical item *just* in general or a more specific *exclusive* function shared with *only*—i.e., do they encode extremely fine-grained sense-similarities, or are they word-specific?. If the latter is true, localization should transfer more strongly between *exclusive just* and *only* than between *non-exclusive just* and *only*.

Methodology Our localization approach follows Experiment 3a, changing only the source and target groups. As described in Section 3.1, the 60 *just* items comprise 30 *exclusive* and 30 *non-exclusive* cases. We construct a separate top-500 positive-advantage response pool for each *just* subset and reuse the *only* pool from Experiment 3a. Let J_E , J_N , and O denote the response pools for *exclusive just*, *non-exclusive just*, and *only*, respectively. We evaluate the four source–target pairings

$$\mathcal{T} = \{J_E \rightarrow O, J_N \rightarrow O, O \rightarrow J_E, O \rightarrow J_N\},$$

where each arrow points from the pool used to localize sites to the pool supplying the evaluation responses.

For each pairing $A \rightarrow B \in \mathcal{T}$, we run both fold directions. We first localize sites on source fold 1 and evaluate them with the necessity intervention on target fold 2 ($A_1 \rightarrow B_2$); we then swap the folds ($A_2 \rightarrow B_1$). We average these two estimates and otherwise use the same component types, values of k , and response aggregation as in Experiment 3a. Consequently, the $J_E \rightarrow O$ and $J_N \rightarrow O$ pairings are evaluated on the same *only* response folds, whereas the $O \rightarrow J_E$ and $O \rightarrow J_N$ pairings use the same *only*-localized sites but different *just* response pools. For each transfer direction, the two rightmost columns of Table 1 report the difference between the *non-exclusive* and *exclusive* necessity effects obtained under localized-site patching:

$$\Delta_{\text{sense}} = \Delta_{\text{patch}}^{\text{non-exclusive}} - \Delta_{\text{patch}}^{\text{exclusive}}.$$

For *just*→*only*, the two terms are the effects from $J_N \rightarrow O$ and $J_E \rightarrow O$; for *only*→*just*, they are the effects from $O \rightarrow J_N$ and $O \rightarrow J_E$. Because a stronger necessity effect corresponds to a more negative Δ_{patch} , positive Δ_{sense} values favor *exclusive just*, whereas negative values favor *non-exclusive just*.

Results Transfer is sense-selective in the *only*→*just* direction, but not in the reverse direction. Every *only*→*just* contrast is positive: sites localized on *only* have a larger necessity effect on *exclusive just* than on *non-exclusive just*. This *exclusive* advantage is significant for residual-stream sites at every k and for the top-10 attention sites, but not for MLP sites. No *just*→*only* contrast shows a significant *exclusive* advantage. Instead, the significant attention contrasts at $k = 1$ and $k = 3$ and the residual-stream contrast at $k = 5$ favor *non-exclusive just*. Thus, *only*-localized sites transfer selectively to the *exclusive* sense of *just*, but sites localized on *exclusive just* do not preferentially transfer back to *only*. This asymmetry supports partial, direction-dependent sharing concentrated in the residual stream rather than a single *exclusive* mechanism that is equally recoverable from either particle.

7 Discussion and Conclusion

In this work, we find LLMs are sensitive to the English discourse particles *just* and *only* with respect to continuations whether using ground-truth follow-ups (Section 4) or LLM-generated follow-ups (Sec-

Component	top- k	Localization: Δ_{patch} ($\downarrow$ = stronger)						Transfer: Δ_{sense} ($+$ = J_E stronger)	
		<i>not</i>		<i>just</i>		<i>only</i>		<i>just</i> $\rightarrow$ <i>only</i>	<i>only</i> $\rightarrow$ <i>just</i>
		localized	random	localized	random	localized	random		
Attn.	1	-0.47	-0.06	-0.22	-0.01	-0.34	-0.12	-0.36	0.01
	3	-2.15	-0.41	-1.09	-0.07	-2.10	-0.13	-0.78	0.18
	5	-6.59	-0.35	-1.72	-0.12	-3.29	-0.12	-0.60	0.52
	10	-15.88	-0.34	-2.98	-0.10	-5.51	-0.18	0.29	1.15
MLP	1	-0.46	-0.04	-0.26	-0.09	-0.38	-0.08	0.10	0.17
	3	-0.84	-0.41	-0.75	-0.08	-0.99	-0.10	0.04	0.25
	5	-1.64	-0.66	-0.87	-0.16	-1.10	-0.19	-0.42	0.27
	10	-3.46	-0.24	-1.11	-0.28	-1.34	-0.24	-0.24	0.52
Resid.	1	-5.31	-0.29	-1.79	-0.49	-3.25	-0.84	0.25	0.84
	3	-7.94	-5.35	-2.23	-0.36	-3.89	-1.87	0.02	1.14
	5	-12.13	-3.67	-2.66	-0.78	-5.15	-1.37	-0.65	1.34
	10	-18.32	-1.53	-3.45	-0.55	-6.82	-0.92	0.00	1.52

Table 1: Component-localization and cross-particle-transfer results ($\times 10^{-2}$, per token) for Llama 3 8B Instruct, by component and number of patched component-layer sites k . **Left:** mean change in log-probability under the necessity intervention, Δ_{patch} . Localized sites are compared with the mean of four equally sized random-site sets. More negative values indicate stronger necessity effects. Boldface indicates a significantly larger necessity effect for localized than random sites ($p < .05$, Bonferroni correction). **Right:** the **non-exclusive**-minus-**exclusive** difference in necessity effects under localized-site patching, Δ_{sense} , for each transfer direction. Positive values indicate a stronger effect for **exclusive** *just*, and negative values indicate a stronger effect for **non-exclusive** *just*. Boldface indicates a significant difference (two-tailed $p < .05$, Bonferroni correction).

tion 5), aligning well with human judgments (Section 4) particularly in a weaker preference for *only* than *just*. This suggests LMs have meaningfully acquired some of the subtlety of these particles’ effects on continuations. Furthermore, when tested according to their own distributions of continuations, LMs continue to display this sensitivity with their own generations (Section 5), aligning with Warstadt (2020)’s theory of *just*. In both these experiments, we find instruction tuning to help increase LLM sensitivity to these particles.

We also find that this sensitivity is localized asymmetrically in our target LLM Llama 3 8B Instruct: Units sensitive to *only* transfer more cleanly to **exclusive** *just* than **non-exclusive** *just*, but units for both sense categories of *just* transfer equally well to *only*. This suggests that, although there is some unified localization for exclusives (as predicted by Coppock and Beaver (2014)), LLMs represent these particles lexically, with senses bundled together, rather than as separate senses of the same lexical item.

We hope this work inspires more investigations into LM sensitivity to pragmatic phenomena that consider upcoming conversational continuations, and provides methodological scaffolding for such investigations. Our data and code are available at (URL).

8 Limitations

This work is limited to English discourse particles in relatively short, scripted-style exchanges; more work is needed for more particles, especially more exclusives, as well as those in other languages. Additionally, although annotator agreement on our stimuli is high and distractor controls behave as expected, the data are ultimately hand-selected by one linguist, which may be particularly biased in the sense labels. This concern is slightly mitigated in Experiment 2 where we generate a distribution of continuations, but it is not completely gotten rid of, since ultimately the prefixes are limited to the selected DailyDialog items.

Concerning our choice of LMs, we only experiment on those in the 8B parameter range. While this size is useful for the level of compute we have access to, both smaller and larger LLMs may behave quite differently. Also, since we analyze only one LLM in experiment 3, more testing is needed to see LLM trends writ-large, and our method may miss structure that exists in finer-grained units such as individual heads or neurons.

Finally, our minimal pairs rely on the counterfactual S' with removal of the target word; while we did our best to choose data where S' was very plausible, subtler fluency artifacts and contextual cues may persist and add noise.

Acknowledgments

Kanishka Misra was supported by a Donald D. Harrington Faculty Fellowship during a part of this project.

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A Prompt Construction Example

We illustrate our prompt construction with one *just* item from DailyDialog. The dataset fields shared by the base and instruction-tuned conditions are:

Context C : Speaker 1: I am sorry for damaging your car, sir.

Original utterance S : Speaker 2: Don't worry about it. Just leave it to the insurance companies.

Modified utterance S' : Speaker 2: Don't worry about it. Leave it to the insurance companies.

Gold follow-up F : Speaker 1: The car park in this community is over crowded, isn't it?

The gold follow-up F is part of the original DailyDialog item. In the generation experiment, it is not included in the prompt; instead, the model samples follow-ups F_{CS}^* and $F_{CS'}^*$ from the original and modified histories, respectively. Below, we show the second unique response sampled from each history under the experiment's fixed seed.

A.1 Base-model rendering

For Llama 3 8B, the two histories are rendered with alternating speaker labels and a final next-speaker prefix:

Original history CS .

Speaker 1: I am sorry for damaging your car, sir.
Speaker 2: Don't worry about it. Just leave it to the insurance companies.
Speaker 1:

Modified history CS' .

Speaker 1: I am sorry for damaging your car, sir.
Speaker 2: Don't worry about it. Leave it to the insurance companies.
Speaker 1:

The corresponding sampled follow-ups are:

F_{CS}^* : Will the insurance companies cover the cost of repair?

$F_{CS'}^*$: No, I want to pay for the damage myself.

A.2 Instruction-tuned rendering

For Llama 3 8B Instruct, we use the checkpoint's native chat template without adding a system prompt. Roles alternate so that S or S' is the final user message and the model generates the next assistant message. Thus, for this item, the single turn in C is assigned the assistant role. The exact serialized prompts are:

Original history CS .

```
<|begin_of_text|><|start_header_id|>assistant<|_j|_>
↪ end_header_id|>
```

```
I am sorry for damaging your car, sir.<|eot_id|_j|_>
↪ ><|start_header_id|>user<|end_header_id|>
```

```
Don't worry about it. Just leave it to the
↪ insurance companies.<|eot_id|><|start_header_id|>assistant<|end_header_id|>
```

Modified history CS' .

```
<|begin_of_text|><|start_header_id|>assistant<|_j|_>
↪ end_header_id|>
```

```
I am sorry for damaging your car, sir.<|eot_id|_j|_>
↪ ><|start_header_id|>user<|end_header_id|>
```

```
Don't worry about it. Leave it to the insurance
↪ companies.<|eot_id|><|start_header_id|>assistant<|end_header_id|>
```

The corresponding sampled follow-ups are:

F_{CS}^* : That's the spirit! Yeah, let the professionals handle it. I'm just glad no one was hurt. Accidents can be a real hassle, but it's good that we're taking responsibility and letting the insurance companies sort it out.

$F_{CS'}^*$: Thank you, sir! I appreciate your understanding. I know accidents can be frustrating, but I'm glad we can handle it through insurance. Is there anything else I can do to make it right?

B Human Annotation Details

Annotators were recruited through Prolific, and compensated at 18USD/hour. Annotators were required to (1) have primary residence in the United States, (2) have English as their primary language, and (3) have an approval rating of greater than or equal to 95%. See Figure 4 for the annotation interface. We changed the slider from continuous to an implicit 7-pt Likert scale based on the feedback from our pilot annotators that the continuous slider made them worry too much about the exact rating for one datum compared to another.

Figure 4: Annotation interface. Annotators were tasked to rate which version of the conversation they preferred by using the slider.

C Additional Particle Removal Results

Tables 2 and 3 contain detailed results for the particle-removal experiments in Section 4.

D LLM Particle-sensitive Generation

Table 4 contains full results for Section 5 as presented in Figure 3.

E Compute and Model Details

LLMs and implementation. The exact Hugging Face checkpoints used in Experiments 1 and 2 are listed in Table 5. Experiment 3 uses only Meta-Llama-3-8B-Instruct, as motivated in Section 6.1. All neural experiments are inference-only: we do not train or fine-tune any model. We run the checkpoints with PyTorch and Hugging Face Transformers in half precision (FP16), with gradient computation disabled. Statistical analyses and figure generation run on CPUs and contribute negligible accelerator compute relative to model inference.

Hardware and accounting. Experiments use a pool of three NVIDIA A40 GPUs with 48 GB of memory per GPU. Each A40 holds one model replica, and independent data shards or Experiment 3 jobs are run in parallel when possible. The estimates below are rounded and include model

Model	Part.	% inc.	Median Δ_{rem}	95% CI	N
Llama 3 8B	just	69.7	0.057	[0.050,0.064]	1121
	only	73.8	0.071	[0.061,0.084]	347
	not	97.1	0.373	[0.358,0.388]	1269
OLMo-2 7B	just	72.0	0.060	[0.054,0.069]	1112
	only	77.1	0.098	[0.079,0.118]	349
	not	98.0	0.415	[0.390,0.433]	1273
Qwen3.5 9B Base	just	66.0	0.044	[0.034, 0.049]	1114
	only	71.6	0.069	[0.050, 0.084]	349
	not	96.6	0.327	[0.318, 0.345]	1276
Gemma-2 9B	just	69.7	0.051	[0.043,0.058]	1110
	only	73.9	0.075	[0.064,0.089]	348
	not	97.7	0.345	[0.331,0.358]	1269

Table 2: Sensitivity to particle removal, measured by the per-token removal delta $\Delta_{\text{rem}} = \log P(CSF) - \log P(CS'F)$. % inc. is the percentage of examples with positive Δ_{rem} . N is the number of non-outlier examples retained after IQR-based outlier filtering.

Model	Part.	% inc.	Median Δ_{rem}	95% CI	N
Llama 3 8B Instruct	just	85.3	0.189	[0.176,0.207]	1112
	only	80.3	0.152	[0.129,0.172]	350
	not	97.3	0.528	[0.502,0.559]	1253
OLMo-2 7B Instruct	just	80.5	0.150	[0.132,0.167]	1136
	only	78.1	0.159	[0.138,0.199]	351
	not	96.7	0.536	[0.511,0.555]	1281
Qwen3.5 9B CT	just	63.5	0.050	[0.039,0.058]	1112
	only	67.8	0.078	[0.053,0.095]	348
	not	95.0	0.394	[0.375,0.411]	1271
Gemma-2 9B IT	just	82.9	0.149	[0.132,0.166]	1131
	only	80.1	0.134	[0.114,0.160]	341
	not	97.3	0.454	[0.429,0.473]	1260

Table 3: Instruction-tuned LLM sensitivity to particle removal, measured by the per-token removal delta $\Delta_{\text{rem}} = \log P(CSF) - \log P(C'S'F)$. % inc. is the percentage of examples with positive Δ_{rem} . N is the number of non-outlier examples retained after IQR filtering.

loading and inference. GPU-hours sum time over occupied accelerators, so three GPUs used for 30 minutes count as 1.5 GPU-hours even when they run concurrently.

Experiment 1. Scoring the 180-item gold dataset in Section 4 takes less than 10 minutes per checkpoint on one A40, or less than 1.3 A40 GPU-hours across all eight checkpoints. Repeating the same scoring procedure on the full filtered set of 2,849 item pairs for the supplemental analysis in Appendix C requires approximately 8 A40 GPU-hours in total; runtime varies substantially across checkpoints because of differences in architecture and prompt serialization. Human annotation and the subsequent correlation and significance analyses require no GPU compute.

Experiment 2. For each checkpoint, we generate 50 responses from each of CS and $C'S'$ for each of the 180 gold items, giving 18,000 sampled responses per checkpoint and 144,000 across all eight checkpoints. Each response is then rescored under both prompt variants. With sampling sharded across three A40s, a checkpoint takes approximately 25–30 minutes of wall time, corresponding to about 1.4 A40 GPU-hours per checkpoint and approximately 11 A40 GPU-hours for the complete experiment.

Experiment 3. The reported component analysis comprises three native particle localiza-

tion/evaluation jobs, two additional jobs for the **exclusive** and **non-exclusive** *just* pools, and four cross-particle transfer jobs. Each job uses one A40, with up to three jobs run concurrently. Individual jobs take approximately 24–57 minutes, and the nine jobs sum to approximately 4.6 A40 GPU-hours. This total includes localization over all 32 layers and three component types, held-out sufficiency and necessity evaluation for generated responses, and the four random-site controls described in Section 6.

Across the model-based experiments, including the supplemental larger-dataset analysis, the reported runs require approximately 25 A40 GPU-hours in total.

F AI Assistant Use

AI assistants were used in this work as coding assistants when implementing experiments and visualizations, to provide suggestions to improve writing flow, and to polish/clarify author-written text. When AI assistant outputs were included in this work, they were thoroughly verified by authors for accuracy.

		not		just		only	
		S	S'	S	S'	S	S'
Llama3	Base	0.0554 ± 0.00242	0.0598 ± 0.00334	0.0266 ± 0.00207	0.0164 ± 0.00165	0.0202 ± 0.00141	0.0161 ± 0.00124
	Instruct	0.207 ± 0.005	0.299 ± 0.00733	0.0297 ± 0.000966	0.0297 ± 0.00117	0.054 ± 0.00175	0.0481 ± 0.00145
OLMo2	Base	0.0402 ± 0.00365	0.084 ± 0.0044	0.0126 ± 0.00168	0.0224 ± 0.00181	0.0231 ± 0.00158	0.0197 ± 0.00152
	Instruct	0.215 ± 0.00508	0.264 ± 0.00561	0.0392 ± 0.00214	0.0324 ± 0.00162	0.0405 ± 0.00185	0.044 ± 0.00168
Gemma2	Base	0.0599 ± 0.00222	0.0659 ± 0.00312	0.0166 ± 0.00106	0.0146 ± 0.00125	0.0189 ± 0.00158	0.0168 ± 0.00112
	Instruct	0.217 ± 0.00966	0.402 ± 0.0105	0.0266 ± 0.00329	0.0558 ± 0.00304	0.0313 ± 0.00371	0.0858 ± 0.00344
Qwen3.5	Base	0.0613 ± 0.00316	0.0986 ± 0.00506	0.0152 ± 0.00158	0.0324 ± 0.00248	0.0221 ± 0.00198	0.0202 ± 0.00167
	Instruct	0.244 ± 0.00475	0.309 ± 0.00553	0.0423 ± 0.00162	0.035 ± 0.00142	0.0516 ± 0.00146	0.0523 ± 0.00147

Table 4: Mean candidate-level own-context advantage per token ($\pm$ standard error), by LLM, particle, and response source (S : with word, S' : word removed.)

Family	Size	Base checkpoint	Instruction-tuned checkpoint
Llama 3	8B	meta-llama/Meta-Llama-3-8B	meta-llama/Meta-Llama-3-8B-Instruct
OLMo-2	7B	allenai/OLMo-2-1124-7B	allenai/OLMo-2-1124-7B-Instruct
Qwen3.5	9B	Qwen/Qwen3.5-9B-Base	Qwen/Qwen3.5-9B
Gemma-2	9B	google/gemma-2-9b	google/gemma-2-9b-it

Table 5: Model checkpoints used in Experiments 1 and 2.